

05_Linear Regression

Say you want to predict the **price of a house** based on its **square footage**.

Square Footage (X)	Price (\$Y)
1000	200000
1500	250000
2000	300000
2200 (Testing)	?

Price = (Slope × Square Footage) + Intercept

$$\text{Price}(y) = mx + c$$

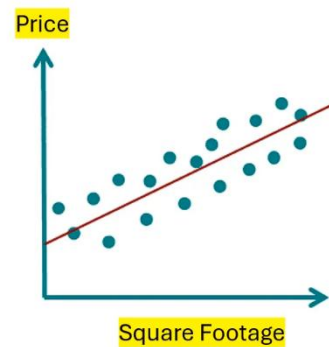
Mean-Based Formula for OLS:

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

$$c = \bar{y} - m\bar{x}$$

$$\text{Price}(y) = mx + c$$

Loss: Original_Price(y) - Price(y)



$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad \left| \quad \text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \left| \quad \text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

The problem with mean based formulas:

In simple linear regression (1 input), you can use:

$$m = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} \quad \text{and} \quad c = \bar{y} - m\bar{x}$$

- r_{12} = correlation between x_1 and x_2
- s_1, s_2 = standard deviations of x_1 and x_2

For two features: $m_1 = \frac{\sum(x_1 - \bar{x}_1)(y - \bar{y}) - r_{12} \cdot s_1/s_2 \cdot \sum(x_2 - \bar{x}_2)(y - \bar{y})}{\sum(x_1 - \bar{x}_1)^2 \cdot (1 - r_{12}^2)}$

For **multiple features**, things get messy:

- You'd need to calculate **partial correlations**, **standard deviations**, and **adjust for multicollinearity** (how features influence each other).
- The formulas for m_1, m_2 become **long and complex**.
- Doesn't scale well to 3, 5, 10, or 100 **features**.
- Easy to make mistakes when features are correlated.

The Normal Equations for Linear Regression:

Use matrix form: $\mathbf{w} = (X^T X)^{-1} X^T y$

X is the matrix of inputs: $X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}$

- First column = all 1s (for the intercept c)
- Each row = one data point
- x_{ij} = value of the j -th feature for the i -th example

Produced Result:

$$\mathbf{w} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$y = m_1 x_1 + m_2 x_2 + c$$

The **first column of 1's** is **not the value of c** . It's a **placeholder column** to learn **C** (the intercept) during matrix multiplication.

These are the values that go into your final regression equation: $y = m_1 x_1 + m_2 x_2 + c$

$$y = m_1 x_1 + m_2 x_2 + \underbrace{c}_{\text{bias/intercept}}$$

Problems Statement:

x_1 (sqft)	x_2 (bedrooms)	y (price)
1000	2	200000
1500	3	250000
2000	4	300000

Let's solve with matrix: $\mathbf{w} = (X^T X)^{-1} X^T y$

$$X = \begin{bmatrix} 1 & 1000 & 2 \\ 1 & 1500 & 3 \\ 1 & 2000 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 200000 \\ 250000 \\ 300000 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1000 & 1500 & 2000 \\ 2 & 3 & 4 \end{bmatrix}$$

Produced Result:

$$\mathbf{w} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$y = m_1 x_1 + m_2 x_2 + c$$